Jinay Patel

jinayunity22@gmail.com | github.com/Github11200 | jinay-patel.vercel.app

Education

University of Waterloo
Bachelor of Computer Engineering *Expected Graduation: Jun 2031*

Skills

Languages: C/C++ (5 years), JavaScript/TypeScript (6 years), Python (5 years), Java (2 years), Rust (3 years)
Technologies: PyTorch, LangChain, Mastra, CMake, vcpkg, React, Svelte, Next JS, Tailwind CSS, Git, Docker, Unity, Linux

Experience

Lead Software Engineer October 2023 — June 2026
Seaquam Robotics *Delta, BC*
- Built a data backend with Deno and TypeScript to aggregate data from 3 sources, cutting manual data entry by 50%
- Created a custom React and Tauri desktop CMS app, reducing data entry time from weeks to 60 minutes
- Built a Deno backend caching layer for slow third-party API endpoints, increasing server **response speed by 150%**

Embedded Controls Software Engineer October 2021 — June 2023
Seaquam Robotics *Delta, BC*
- Implemented C++ motion algorithms, securing a **100% win rate** during autonomous periods across 15+ matches
- Designed an interpolated lookup table using a monotonic cubic spline to tune PID constants in real time, increasing movement accuracy by 200% with **error tolerances under 1 inch**

Software Engineering Tutor September 2025 — March 2026
Code Ninjas *Surrey, BC*
- Helped 5 senior Black Belt students debug and implement features for their games being built with C# and Unity
- Designed ad campaigns with visual assets for Code Ninjas, improving the local branch's search visibility

Freelance Systems Administrator / IT Consultant
Convergence Media *Remote*
- Created a custom Linux Mint ISO using Cubic with 3 bash scripts to automate the installation and setup of programs
- Deployed a self-hosted Nextcloud server (Microsoft 365 alternative) using Docker, which allowed file sharing and private video calls
- Configured a **Tailscale tunnel** to securely expose the Nextcloud server for remote video calls and file sharing

Projects

Motion Control Framework | Embedded Controls Software Engineer GitHub
- Used C++ to enable autonomous robot movement using PID controllers, feedforward controllers, and 2D motion profiles
- Designed a custom **2D motion profile generator** using cubic Bezier curves to calculate the robot's position, velocity, and acceleration at each point on the path while compensating for centripetal forces to keep the robot on the path
- Built **real-time visualization software** with Svelte and WebSockets to display live position, PID data, and logs

Verbosity: Custom Programming Language | Personal Project GitHub, Blog Post
- Built a custom lexer, parser, and code generator from scratch for a custom programming language using C++ and CMake
- Applied C++26 features such as abstract classes, virtual methods, smart pointers, and variants for the compiler pipeline

Termicord - Real-Time CLI Chat Application | Personal Project GitHub
- Developed a Discord-like chat application which runs in the terminal using C++, raw TCP sockets, NCurses (for the TUI), and vcpkg
- Implemented a **custom HTTP server** with routing to handle cURL requests sent to a REST API along with a custom database

Awards

Vex Robotics World Champion, RECF & Vex Robotics April 2026
Top overall team among 20,000+ worldwide, recognized for excellence across robot design, programming, and competitive performance

Excellence Award @ Provincial Championship, Vex Robotics February 2026
Awarded to the top team out of the 100+ teams in British Columbia for excellent performance

VSHacks Hackathon 2nd Place, VSHacks July 2025
Awarded for developing Posturai, a real-time posture correction app

Robotics Programming Award x3, Waterloo CEMC May 2026
Awarded for innovative motion control algorithms and technical documentation

Senior Computer Programming Award, Seaquam Secondary School June 2026